

Mohak A. Prakash

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Education

University of California, San Diego

Expected March 2027

- **B.S. in Data Science, Minor in Cognitive Science (Neuroscience)**
- GPA: 3.8/4.00, Provost Honors (Winter 2024 - Present)
- **Concentrations:** Machine Learning and Artificial Intelligence & Cognitive Neuroscience
- **Relevant Coursework:** DSA, Machine Learning and Probabilistic Modeling, Statistical Methods, Database Management, Linear Algebra, Statistical Inference and Prediction, Empirical Risk Minimization, System Design and Scalable Analytics, Reinforcement Learning, Recommender Systems

Skills

Python, Java, JavaScript, React, Flutter, FastAPI, Flask, PostgreSQL, MongoDB, Firebase, Cloud (AWS, GCP), Docker, Kubernetes, Pandas, Cvxpy, Tensorflow, PyTorch, Sklearn, Clustering (K-means, DBSCAN, GMM), SVM, Random Forests, LDA/QDA, Q-Learning, Transformers, NLP (NER, Relation Extraction), LangChain, LangGraph, LlamaIndex, MCP, LLM and VLM (OpenAI, Claude, Gemini), RAG, Knowledge Graphs, OCR, OpenCV, MediaPipe, Twilio, Ngrok

Work Experience

Research Assistant — *Proto Lab, UC San Diego Design Lab*

Sep 2025 - Present

- Built an MVP of an Agentic meeting simulator where multiple AI agents engage in multi-round discussions, dynamically exchanging ideas based on a configurable agreeability setting to generate more insightful outcomes.

Research Intern — *EXL Services*

Jun 2025 - Sep 2025

- Contributing to development of RiskXpert, an Agentic AI tool for automated insurance and risk analysis, focusing on risk assessment frameworks and control identification methodologies.
- Built multiple OCR evaluation pipeline exploring AWS Textract, DocLLM, PaddleOCR+LayoutLMv3, and a custom CNN+RNN+CRF handwriting stack to benchmark performance across diverse document structures.
- Applied ML, compliance (SOC 2 Type 2, GDPR), data security principles, monetary and computational cost to evaluate each approach and determine the most effective, regulation-aligned solution.

Research Assistant — *Knight Lab, University of California San Diego*

Jan 2025 - Present

- Built biomedical knowledge graph pipeline processing 2000+ research papers using BioBERT NER and custom PubMedBERT relation extraction model. Got 70% F1-score on MicrobioRel dataset (matching 71.3% benchmark).
- Integrated Llama 3 8B and Mistral 7B for text preprocessing and entity extraction, implementing chunk-based processing to optimize BERT context handling across 22 relation classes.

Projects

Clueless — [Project Link](#)

Winner @ HackMIT '25

- Engineered automated knowledge graph construction of web applications by extracting DOM hierarchies, clickable elements, and semantic relationships into Neo4j; implemented BFS-based traversal for shortest-path workflow guidance across dynamic UIs, with synchronized frontend overlays in a React/TypeScript Plasmio extension.
- Built a multi-agent AI backend where Claude handles query classification and intent extraction, Pinecone + Sentence Transformers enable semantic RAG, and Cerebras accelerates multi-step reasoning, enabling real-time navigation, retrieval, and cross-application developer onboarding.

TaxDaddy — [Project Link](#)

Winner @ Hacklytics '25

- Engineered a Real-Time Voice Agent with WebSockets, Ngrok and Twilio integration capable of Real-Time RAG and multi-level tool calling. Agent accesses tax knowledge from a Pinecone VectorDB, coordinates document completion through a Form Agent, and orchestrates data between components with a multi-agent graph workflow.
- Developed an intelligent document processing system using OCR to extract information into MongoDB, enabling automatic tax form identification and completion with PyPDF, and secure delivery via integrated email services

Achievements and Activities (3x Hackathon winner)

- 2025: Screening Track Winner @ HealthLink Hackathon
- 2025: Second Place, Wispr Track @ HackMIT
- 2025: Third Overall, Hacklytics @GT
- **Department Tutor** @ HDSI. (DSC40A, DSC20)
- **Triton NeuroTech Project Member** (since W125)
- **Founding Member of Khoj** (Indie-Fusion Music Club)